

ANWAR UDIN SAYFULLOH

AI Engineer | Machine Learning | Geoscience Data

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PROFESSIONAL SUMMARY

AI-focused engineer with a geophysics degree from Institut Teknologi Sumatera (ITERA) and hands-on experience building end-to-end machine learning and deep learning solutions — from data preprocessing and feature engineering to model deployment. Published researcher in geophysical inversion and uncertainty analysis (Python, Numba, Joblib). Former Marine Geophysics Intern at PT Qyudos Geosurvey and AI Data Annotator at Sahara AI. Currently completing the AI Engineering program at Purwadhika Digital Technology School; seeking AI Engineer / ML Engineer roles.

TECHNICAL SKILLS

Programming: Python, SQL, C++, JavaScript, HTML/CSS

Machine Learning: Scikit-learn, XGBoost, Random Forest, SMOTE, GridSearchCV, feature engineering, classification, regression, clustering (K-Means, DBSCAN), PCA

Deep Learning: PyTorch, TensorFlow/Keras, CNN, ResNet34, EfficientNet, YOLO, transfer learning, data augmentation

LLM & NLP: RAG, prompt engineering, LLM evaluation, data annotation for LLM training

Data & Analytics: Pandas, NumPy, SciPy, EDA, data cleaning, data visualization, statistical analysis

MLOps & Deployment: FastAPI, Docker, MLflow, Streamlit, Git/GitHub, Jupyter

Geoscience: Magnetotelluric (MT), forward/inverse modeling, uncertainty analysis, seismic acquisition & processing, GPR, ERT/VES, petrophysics, well correlation, QGIS, Petrel, Oasis Montaj

Tools: OpenCV, Numba, Joblib, MATLAB, SonarWiz, Surfer, Microsoft Office

Languages: Indonesian (native), English (working proficiency)

PROFESSIONAL EXPERIENCE

Marine Geophysics Intern — PT Qyudos Geosurvey Indonesia | Jul 2025 – Aug 2025 | Bandung, Indonesia

- Processed Sub-Bottom Profiler (SBP) data for subsea pipeline route analysis, identifying sedimentary layers and buried hazards to support engineering decisions.
- Conducted Side Scan Sonar (SSS) acquisition and interpretation for seafloor imaging; performed real-time interpretation during acquisition.
- Assisted with geophysical equipment calibration and acquisition quality control; used SonarWiz and Golden Software Surfer for processing and interpretation.

AI Data Annotator (Web3 AI) — Sahara AI | Dec 2024 – May 2025 | Remote

- Annotated and reviewed image, text, and video datasets per project guidelines to support AI model development.
- Performed data quality verification to identify inconsistencies and improve annotation accuracy.
- Conducted AI model robustness testing, including prompt manipulation and jailbreak testing; evaluated AI-generated outputs for consistency.

Team Lead / Technical Lead — GARAGA Team, ANTAM Hackathon (Unit Geomin) | 2026

- Led the technical team in developing a data-driven solution for an exploration-related problem.
- Coordinated task allocation, technical discussions, and submission deliverables; integrated team analysis into a coherent solution.

SELECTED PROJECTS

1D Magnetotelluric Inversion & Uncertainty Analysis (Python, GUI + CLI)

Link: github.com/ipul122/parallel-neighborhood-algorithm

- Implemented Sambridge's Neighborhood Algorithm for 1D MT forward modeling, inversion, and uncertainty quantification.
- Improved computational performance with Numba JIT compilation and Joblib multi-core parallelization; supports .edi field data.

Gold Deposit Mapping Probability & Spatial Uncertainty (ML + GIS)

Link: github.com/ipul122/Gold-Deposit-Mapping

- Built a mineral prospectivity workflow integrating radiometric, geological, geochemical, and geophysical evidence with Scikit-learn, XGBoost, and QGIS geospatial processing.

Rock Classification Using Deep Learning (PyTorch + Streamlit)

Link: github.com/ipul122/rock-classification

- Developed image classification models with a custom CNN and ResNet34 transfer learning; OpenCV preprocessing; interactive Streamlit demo app.

Subsurface Utility Detection on GPR (YOLO Computer Vision)

Link: github.com/ipul122/land-risk-prediction-using-machine_learning

- Built an automated pipeline detecting GPR hyperbola reflections from underground utilities; YOLO with custom augmentation across varied scan resolutions.

Landslide Risk Prediction (Multiclass ML)

Link: github.com/ipul122/land-risk-prediction-using-machine_learning

- End-to-end workflow with StandardScaler, SMOTE, XGBoost, and Random Forest; GridSearchCV hyperparameter tuning; evaluated with accuracy, precision, recall, and F1-score.

PUBLICATION

Sayfulloh, A. U. (2026). Uncertainty Analysis of 1D Magnetotelluric Inversion Model Using the Neighborhood Algorithm: Application to Kilauea Volcano. Available at researchgate.net/profile/Anwar-Sayfulloh/research

EDUCATION

B.Eng (S.T.), Geophysics — GPA: 3.31/4.00 — Institut Teknologi Sumatera (ITERA) | 2022 – 2026 | Lampung, Indonesia

Thesis: 1D Magnetotelluric (MT) Data Inversion and Uncertainty Analysis Using Sambridge's Neighborhood Algorithm.
Coursework: Exploration Geology, Geophysical Methods, Geophysical Data Processing, Geophysical Modeling, Remote Sensing.

AI Engineering Program — Purwadhika Digital Technology School | Aug 2026 – Present

Practical training in Python, data processing, machine learning, SQL, and AI engineering with software engineering best practices.

FIELDWORK

- Geophysical Field Surveys (ITERA & Kab. Tanggamus): gravity, magnetic, seismic, GPR, and geoelectrical (VES/ERT) acquisition and processing; operated Scintrex CG-6 gravimeter, magnetometer, SummitX seismic system, and GPR equipment.
- Geological Field Courses (Lampung): sedimentology, stratigraphy, mineralogy, petrology, structural geology; rock/mineral identification and structural measurements (strike & dip).